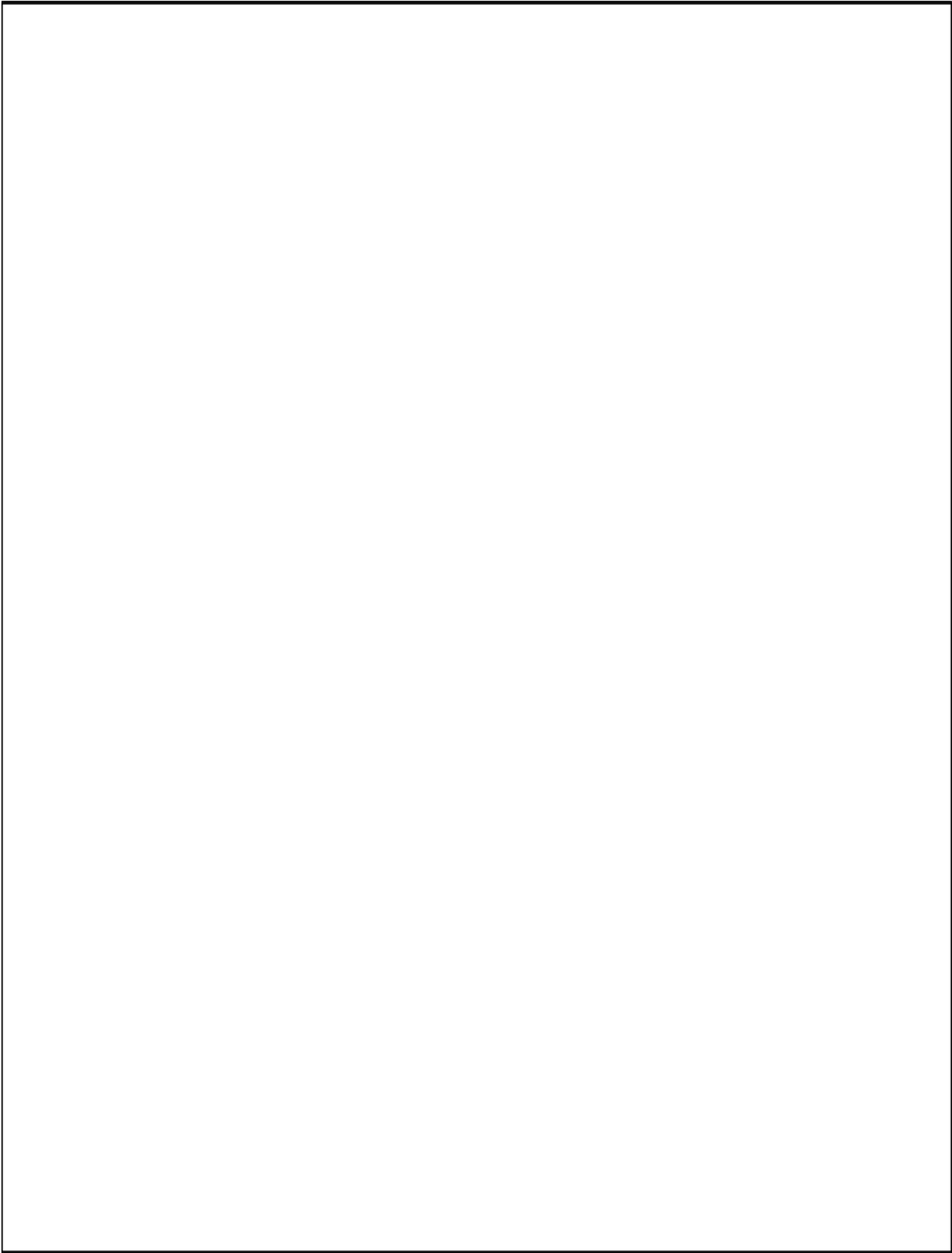


PR. 3 Collection Manipulator



Project: Collection Manipulator

Objective:

Create a Python program called Student Data Organizer that manages a collection of student records. This project will apply intermediate-level concepts such as string formatting and manipulation, collection data types (List, Tuple, Set, and Dictionary), mutability and immutability, type casting, and the del keyword.

Requirements:

String Formatting and Manipulation:

- Gather and store student information (e.g., name, age, grade, and subjects) using input and format the information for display in a user-friendly way.
- Demonstrate different string formatting methods (e.g., f-strings, .format(), or % formatting) to format student details.

Collection Data Types:

- Use a List to store multiple student records and their information.
- Use a Tuple to store each student's unique and unchangeable information (e.g., student ID and date of birth).
- Use Set to manage and display unique subjects offered to students (ensuring no duplicates).
- Use Dictionary to organize student data where keys are student IDs, and values are dictionaries containing the student's name, age, grade, and list of subjects.

Mutability and Immutability:

- Demonstrate the mutability of List by adding or modifying student details.
- Use the immutability of Tuple to handle each student's ID and date of birth, which should not change once set.

Type Casting and del Keyword:

- Perform type casting where needed, such as converting user input strings to integers or floats when required.
- Use the del keyword to delete a student record from the main list based on student ID.

Program Flow:

Welcome and Instructions:

- Display a welcome message and an overview of the program's functionality.

Student Data Collection:

- Prompt the user to enter details for each student: name, age, grade, subjects (as a comma-separated string), student ID, and date of birth.
- Store the student's ID and date of birth as a tuple, add subjects to a set, and store each student's information in a dictionary. Add this dictionary to a list of all student records.

Menu-Driven Options:

- Add Student: Add a new student record.
- Display All Students: This option displays all student records using formatted output, listing the student

ID, name, age, grade, and subjects.

- Update Student Information: Update specific mutable information, like age or subjects, of a selected student by ID.
- Delete Student: Remove a student record using student ID using del.
- Display Subjects Offered: Display all unique subjects offered by students, using a set to ensure no duplicates.

Exit Message:

- Thank the user for using the Student Data Organizer and displaying an exit message.

Example Console Interaction:

Welcome to the Student Data Organizer!

Select an option:

1. Add Student
2. Display All Students
3. Update Student Information
4. Delete Student
5. Display Subjects Offered
6. Exit

Enter your choice: 1

Enter student details:

Student ID: 101

Name: Alice

Age: 20

Grade: B+

Date of Birth (YYYY-MM-DD): 2002-05-14

Subjects (comma-separated): Math, Science, English

Student added successfully!

Select an option:

1. Add Student
2. Display All Students

...

--- Display All Students ---

Student ID: 101 | Name: Alice | Age: 20 | Grade: B+ | Subjects: Math, Science, English

...

Select an option:

...

Instructions:

Task Completion: Ensure that you attempt all the assigned tasks given to you as part of the exam. Complete each task to the best of your ability, following the instructions provided.

Assumptions: Make suitable assumptions wherever necessary, based on the requirements and instructions provided. Document any assumptions made in your project documentation or README.md file.

GitHub Repository: Create a GitHub repository to host your project. Upload your project files, including source code, and documentation to the repository. Ensure that you provide a clear and descriptive README.md file.

No Copying: Do not copy code or any other content from your classmates or any other source. Plagiarism is strictly prohibited and can result in severe consequences, including academic penalties. Ensure that all the code and content in your project are original and properly attributed to the appropriate sources, if applicable.

Submission: Once you have completed your project and uploaded it to your GitHub repository, submit the GitHub repository link to your instructor or as instructed. Double-check that your repository is properly organized and includes all the required files as per the instructions provided.

Remember to follow the instructions provided professionally, make suitable assumptions wherever necessary, and avoid copying code or content from unauthorized sources. Good luck with your project work!

Collection Manipulator

PYTHON - DATA SCIENCE

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“scaling” higher....!!!

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Shaping “skills” for